Mohammad Mohammad

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EDUCATION

Carleton University, Ottawa, ON *Expected May 2028*

B.C.S. Honours in Computer Science, AI & Machine Learning Stream | GPA: 3.92/4.0

- Dean's Honours List; coursework: Data Structures & Algorithms, Software Engineering, Operating Systems

EXPERIENCE

QA Automation Developer *May 2026 – Present*

Government of Canada, Department of National Defence | Ottawa, ON

- Designed and wrote QASE, a C# test-automation framework that converts recorded user interactions into Excel-driven Selenium scripts, so testers no longer write each case by hand.
- Kept parsing, execution and reporting as separate components, so new features plug in without changes to the core.
- Built a debug mode that pauses a failed run, lets the tester edit the Excel steps live and resumes from the last passing step; wrote the specs for dashboarding, test tagging and CI/CD integration.

IP Network Engineering Intern *Jan 2026 – Apr 2026*

Nokia | Ottawa, ON

- Traced data-path performance problems under high-throughput load with packet-level analysis tools and worked with firmware engineers to tune the system.
- Wrote Python and Tcl regression suites that test protocol state machines and security logic.
- Built a Python testbed-mapping tool that automated environment setup, so regression runs start from the same state.

PROJECTS

TitanInfer: Neural-Network Inference Engine *Feb 2026 – Mar 2026*

C++20, AVX2, CMake, GoogleTest, GitHub Actions, Docker, gRPC | Personal project

- C++20 inference library with no BLAS dependency and 32-byte-aligned RAII tensors; wrote and tested a standalone cache-blocked AVX2/FMA matmul kernel.
- Designed a versioned binary model format with a PyTorch exporter. The engine pre-allocates per-layer buffers at load, profiles each layer, and exposes a C API over a mutex-guarded handle.
- Extended it into a multi-tenant model server (LRU eviction, quotas, versioned routes) and a cluster scheduler with power-of-two-choices routing and per-node circuit breakers.
- Tested with 346 GoogleTest cases in GitHub Actions on Ubuntu and Windows, plus a Docker chaos test.

LinguaSpy: Language-Learning Spy Game *Sept 2026 – Present*

Unreal Engine 5 (C++), Python, Blender | Personal project

- Built a top-down language-learning game in Unreal Engine 5 with ~14K lines of C++: 4 cities, 6 missions, a Leitner spaced-repetition scheduler and an accent-tolerant sentence grader.
- Versioned the save format with stepwise migrations. Old files loaded as current because UE skips writing values equal to the class default; the version field now defaults to 0 and is stamped on save.
- Wrote ~4K lines of Python tooling: editor scripts that build all dialogue and missions, and offline TTS voice builds keyed by SHA-1 so only changed lines regenerate.
- 23 headless UE automation tests cover the C++ core; a Slate editor tab edits and validates dialogue graphs with undo.

Monte Carlo Path Tracer *Oct 2025, Nov 2026*

C++17, OpenMP, GLFW/OpenGL, WebAssembly | Personal project, built from the Ray Tracing in One Weekend series

- Path tracer with a binned-SAH BVH (~250x faster on 17K triangles) and OpenMP; fixed five sampling and shading bugs.

TECHNICAL SKILLS

Languages: C++ (C++17/20), C#, Python, JavaScript, Tcl

Testing & CI: Selenium, GoogleTest, Unreal Automation tests, GitHub Actions

Tools & Platforms: Git, CMake, Docker, gRPC, Linux, Visual Studio, Unreal Engine 5, Blender Python API